TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

012900 **PAYMENT PROCEDURES**

013000 **ADMINISTRATIVE REQUIREMENTS**

DIVISION 02 - EXISTING CONDITIONS

024100 **DEMOLITION**

028319 **LEAD BASED PAINT ENCAPSULATION**

DIVISION 07 - THERMAL AND MOISTURE PROTECTION

073100 **SYNTHETIC ROOF TILES**

DIVISION 08 - OPENINGS

081113 **HOLLOW METAL DOORS AND FRAMES**

081416 **FLUSH WOOD DOORS**

087100 **DOOR HARDWARE**

DIVISION 09 - FINISHES

099000 **PAINTING AND COATING - COMMERCIAL FACILITY GUIDE SPECIFICATION - SHERWIN-WILLIAMS**

DIVISION 12 - FURNISHINGS

123530 **RESIDENTIAL CASEWORK**

DIVISION 22 - PLUMBING

221005 **PLUMBING PIPING**

DIVISION 26 - ELECTRICAL

263100 **PHOTOVOLTAIC COLLECTORS**

SECTION 012900
PAYMENT PROCEDURES

SECTION 012900 - PAYMENT PROCEDURES

PART 1 GENERAL

2.01 SUMMARY

A. Section includes administrative and procedural requirements necessary to prepare and process Applications for Payment.

2.02 SCHEDULE OF VALUES

A. Coordination: Coordinate preparation of the schedule of values with preparation of construction schedule.
   1. Coordinate line items in the schedule of values with items required to be indicated as separate activities in Contractor's construction schedule.
   2. Submit the schedule of values to Government within 14 calendar days of receiving award.
   3. An acceptable schedule of values shall be agreed upon by the contractor and the Government before the first progress payment is processed.

B. Format and Content: Use Project Manual table of contents as a guide to establish line items for the schedule of values. Provide at least one line item for each Specification Section.
   1. Arrange schedule of values consistent with format of AIA Document G703.
   2. Provide a breakdown of the Contract Sum in enough detail to facilitate continued evaluation of Applications for Payment and progress reports.
   3. Closeout Costs. Include separate line items under Contractor and principal subcontracts for Project closeout requirements.
   4. Schedule of Values Revisions: Revise the schedule of values when Change Orders or Construction Change Directives result in a change in the Contract Sum. Include at least one separate line item for each Change Order and Construction Change Directive.

2.03 APPLICATIONS FOR PAYMENT

A. Each Application for Payment following the initial Application for Payment shall be consistent with previous applications and payments as certified by the Government.

B. Payment Application Times: Progress payments are accepted. The period of construction work covered by each Application for Payment shall be listed on the invoice. Coordinate with Government to determine frequency of payments.
   1. Submit draft copy of Application for Payment to Government for review before uploading into the Government Invoice Payment processing platform.

C. Application for Payment Forms: Use AIA Document G702 and AIA Document G703 as a reference form for Applications for Payment.

D. Transmittal: Submit one signed electronic copy of each Application for Payment to the Government via email.
   1. Transmit each copy with a transmittal form listing attachments and recording appropriate information about application.
   2. Include the following with the application: Construction progress schedule & partial release of liens from major subcontractors and vendors.

E. Initial Application for Payment: Administrative actions and submittals that must precede or coincide with submittal of first Application for Payment include the following:
   1. List of subcontractors.
   2. Schedule of values.
   3. Contractor's construction schedule (preliminary if not final).
   4. Products list (preliminary if not final).
   5. Schedule of unit prices.
   6. Submittal schedule (preliminary if not final).
   7. List of Contractor's staff assignments.

8. List of Contractor's principal consultants.
9. Report of preconstruction conference.
10. Certificates of insurance and insurance policies.
11. Performance and payment bonds.

F. Application for Payment at Substantial Completion: After Government issues the Certificate of Substantial Completion, submit an Application for Payment showing 100 percent completion for portion of the Work claimed as substantially complete.
   1. Include documentation supporting claim that the Work is substantially complete and a statement showing an accounting of changes to the Contract Sum.
   2. This application shall reflect Certificate(s) of Substantial Completion issued previously for Government occupancy of designated portions of the Work.

G. Final Payment Application: After completing Project closeout requirements, submit final Application for Payment with releases and supporting documentation not previously submitted and accepted, including, but not limited, to the following:
   1. Evidence of completion of Project closeout requirements.
   2. Insurance certificates for products and completed operations where required and proof that taxes, fees, and similar obligations were paid.
   3. Updated final statement, accounting for final changes to the Contract Sum.
   4. Reference AIA Document G706, G706A, G707.
   5. Evidence that claims have been settled.
   6. Final meter readings for utilities, a measured record of stored fuel, and similar data as of date of Substantial Completion or when Government took possession of and assumed responsibility for corresponding elements of the Work.
   7. Final liquidated damages settlement statement.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION

SECTION 013000
ADMINISTRATIVE REQUIREMENTS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Preconstruction meeting.

B. Construction progress schedule.

C. Submittals for review, information, and project closeout.

D. Requests for Information (RFI) procedures.

E. Submittal procedures.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 PRECONSTRUCTION MEETING

A. Schedule meeting after Notice of Award.

B. Owner will schedule a meeting after Notice of Award.

C. Attendance Required:
   1. Government.
   2. Contractor.

D. Agenda:
   1. Execution of Owner-Contractor Agreement.
   2. Submission of executed bonds and insurance certificates.
   3. Submission of list of subcontractors, list of products, schedule of values, and progress schedule.
   4. Designation of personnel representing the parties to Contract.
   5. Procedures and processing of field decisions, submittals, substitutions, applications for payments, proposal request, Change Orders, and Contract closeout procedures.
   6. Scheduling.

3.02 CONSTRUCTION PROGRESS SCHEDULE

A. If preliminary schedule requires revision after review, submit revised schedule within 5 days.

B. Submit updated schedule with each Application for Payment.

3.03 REQUESTS FOR INFORMATION (RFI)

A. Definition: A request seeking one of the following:

B. Preparation: Prepare an RFI immediately upon discovery of a need for interpretation of Contract Documents. Failure to submit a RFI in a timely manner is not a legitimate cause for claiming additional costs or delays in execution of the work.
   1. Prepare a separate RFI for each specific item.
   2. Prepare in a format and with content acceptable to Owner.
   3. Combine RFI and its attachments into a single electronic file. PDF format is preferred.

C. Content: Include identifiers necessary for tracking the status of each RFI, and information necessary to provide an actionable response.
   1. Official Project name and number, and any additional required identifiers established in Contract Documents.
   2. Discrete and consecutive RFI number, and descriptive subject/title.
   3. Issue date, and requested reply date.
   4. Reference to particular Contract Document(s) requiring additional information/interpretation. Identify pertinent drawing and detail number and/or specification section number, title, and paragraph(s).

5. Contractor's suggested resolution: A written and/or a graphic solution, to scale, is required in cases where clarification of coordination issues is involved, for example; routing, clearances, and/or specific locations of work shown diagrammatically in Contract Documents. If applicable, state the likely impact of the suggested resolution on Contract Time or the Contract Sum.

D. Attachments: Include sketches, coordination drawings, descriptions, photos, submittals, and other information necessary to substantiate the reason for the request.

E. Review Time: Government will respond and return RFIs to Contractor within 14 calendar days of receipt. For the purpose of establishing the start of the mandated response period, RFIs received after 12:00 noon will be considered as having been received on the following regular working day.

F. Responses: Content of answered RFIs will not constitute in any manner a directive or authorization to perform extra work or delay the project. If in Contractor's belief it is likely to lead to a change to Contract Sum or Contract Time, promptly issue a notice to this effect, and follow up with an appropriate Change Order request to COR.

3.04 SUBMITTALS FOR REVIEW

A. When the following are specified in individual sections, submit them for review:
   1. Product data.
   2. Shop drawings.
   3. Samples for selection.
   4. Samples for verification.

B. Submit to Government for review for the limited purpose of checking for compliance with information given and the design concept expressed in Contract Documents.

C. Samples will be reviewed for aesthetic, color, or finish selection.

3.05 SUBMITTALS FOR INFORMATION

A. When the following are specified in individual sections, submit them for information:
   1. Design data.
   2. Test reports.
   3. Manufacturer's instructions.
   4. Other types indicated.

3.06 NUMBER OF COPIES OF SUBMITTALS

A. Electronic Documents: Submit one electronic copy in PDF format; an electronically-marked up file will be returned. Create PDFs at native size and right-side up; illegible files will be rejected.

3.07 SUBMITTAL PROCEDURES

A. General Requirements:
   1. Use a single transmittal for related items.
   2. Submit separate packages of submittals for review and submittals for information, when included in the same specification section.
   3. Transmit using any one of the following forms.
      a. Use Form AIA G810.
      b. Use Form CSI/CSC Form 12.1A.
      c. Use Contractor's form, subject to prior approval by Government.
   4. Apply Contractor's stamp, signed or initialed certifying that review, approval, verification of products required, field dimensions, adjacent construction work, and coordination of information is in accordance with the requirements of the work and Contract Documents.
   5. Schedule submittals to expedite the Project, and coordinate submission of related items.
      a. For each submittal for review, allow 15 days excluding delivery time to and from the Contractor.
      b. For sequential reviews involving Architect's consultants, Owner, or another affected party, allow an additional 7 days.

3.08 SUBMITTAL REVIEW

A. Submittals for Review: Government will review each submittal, and approve, or take other appropriate action.

B. Submittals for Information: Government will not acknowledge receipt, and take no other action.

C. Government's actions will be reflected by marking each returned submittal using virtual stamp on electronic submittals.
   1. Notations may be made directly on submitted items and/or listed on appended Submittal Review cover sheet.

END OF SECTION

SECTION 024100
DEMOLITION

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Building demolition excluding removal of hazardous materials and toxic substances.

B. Selective demolition of building elements for alteration purposes.

C. Abandonment and removal of existing utilities and utility structures.

1.02 RELATED REQUIREMENTS

A. Section 02 83 19 - Lead-Based Paint Encapsulation:

1.03 REFERENCE STANDARDS

A. 29 CFR 1926 - Safety and Health Regulations for Construction; Current Edition.

B. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2022, with Errata (2021).

1.04 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Site Plan: Indicate:
   1. Vegetation to be protected.

C. Demolition Plan: Submit demolition plan as required by OSHA and local AHJs.
   1. Indicate extent of demolition, removal sequencing, bracing and shoring, and location and construction of barricades and fences.
   2. Summary of safety procedures.
   3. Demolition firm qualifications.

PART 2 EXECUTION

2.01 DEMOLITION

A. Remove items as shown on Government provided drawing set. .

B. Remove other items indicated, for salvage, relocation, and recycling.

2.02 GENERAL PROCEDURES AND PROJECT CONDITIONS

A. Do not begin removal until receipt of notification to proceed from Owner.

B. Protect existing structures and other elements to remain in place and not removed.

C. Minimize production of dust due to demolition operations. Do not use water if that will result in ice, flooding, sedimentation of public waterways or storm sewers, or other pollution.

2.03 EXISTING UTILITIES

A. Protect existing utilities to remain from damage.

B. Remove exposed piping, valves, meters, equipment, supports, and foundations of disconnected and abandoned utilities.

2.04 SELECTIVE DEMOLITION FOR ALTERATIONS

A. Existing construction and utilities indicated on drawings are based on casual field observation only.
   1. Verify construction and utility arrangements are as indicated.
   2. Report discrepancies to Architect before disturbing existing installation.
   3. Beginning of demolition work constitutes acceptance of existing conditions that would be apparent upon examination prior to starting demolition.

B. Remove existing work as indicated and required to accomplish new work.
   1. Remove items indicated on drawings.

C. Protect existing work to remain.
    1. Prevent movement of structure. Provide shoring and bracing as required.
    2. Perform cutting to accomplish removal work neatly and as specified for cutting new work.
    3. Repair adjacent construction and finishes damaged during removal work.
    4. Patch to match new work.

2.05 DEBRIS AND WASTE REMOVAL

A. Remove debris, junk, and trash from site.

B. Leave site in clean condition, ready for subsequent work.

C. Clean up spillage and wind-blown debris from public and private lands.

END OF SECTION

SECTION 028319
LEAD BASED PAINT ENCAPSULATION

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Removal and legal disposal of lead-based paint (LBP).

B. Encapsulation of lead-based paint

1.02 REFERENCE STANDARDS

A. Federal, State and local laws and latest guidelines and regulations published by Housing and Urban Development (HUD), Environmental Protection Agency (EPA) and Occupational Safety and Health Agency (OSHA) including but not limited to:
   1. 29 CFR 1910.1025 Lead Standard for General Industry
   2. 29 CFR 1926.62 Construction Industry Lead Standard
   3. 40 CFR Part 745 Requirements for Lead-Based Paint Activities in Target Housing and Child-Occupied Facilities
   4. HUD Guidelines for the Evaluation and Control of Lead-Based Paint Hazards in Housing
   5. Uniform Building Code, Latest Edition
   6. Title 8 Section 1532.1 Cal/OSHA Construction Safety Orders, Lead
   7. Title 17, California Code Of Regulations, Division 1, Chapter 8

1.03 SUBMITTALS

A. See section 013000 – Administrative Requirements for submittal procedures.

B. Training Certificates – Required for all workers performing lead related construction activities. Shall include documentation on a lead training course approved by the California California Department of Health Services (DHS), Lead Accreditation and Certification Unit. Workers shall have current state lead certification and at least one on-site supervisor shall have current certification as Supervisor/Monitor. Certificate must be valid for the duration of work being performed.

C. Proof of insurance for lead paint abatement work to a minimum of $500,000.00 single limit for bodily injury and property damage limits of $1,000,000.00 per occurrence.

D. Abatement Plan: Submit abatement plan to Government for review. Shall include at a minimum:
   1. Material and equipment to be used.
   2. Procedures for work area isolation. Including but not limited to as applicable:
      a. Area containment
   3. Methods and equipment for providing negative air pressure
   4. Negative air pressure monitoring

E. Airlock/clean room entrance/exit

F. Procedures for worker decontamination system, including OSHA compliant decontamination enclosures.

G. Procedures and protocols for lead based paint removal and subsequent encapsulation.

H. Waste handling protocols for loading, hauling and disposal of lead-based paint/lead-based paint containing materials. Including the name of the waste transporter and the waste disposal means and destination location.

I. Written Hazard Communication Plan and SDS for products brought on-site by the Contractor.

J. Written Respiratory Protection Plan

K. Project Work Schedule

L. Health and Safety Plan

M. Current lead-based paint abatement certification and licensing for all personnel performing work under this contract.

N. Final Abatement Report: Submit after work has been completed for record. Shall include:

O. Detailed description and documentation of the site-specific operations performed. Including, but not limited to lead based paint identification, sampling and analytical data, and lead-based quantities removed and disposed.

P. Lead-based paint abatement Contractor Daily Logs.

Q. OSHA Monitoring results; and applicable project -specific data.

R. Before and after site photos

S. Final disposition of the lead-based paint waste including copies of all waste disposal forms

PART 2 EXECUTION

2.01 SITE ACCESS

A. A person may not enter or remain on site at any time during a lead demolition project unless that person is:
   1. The Contractor engaged for the lead abatement project and his employees;
   2. State, County or local enforcement official or his designee;
   3. A Federal, State or local official, or his designee, engaged in research on buildings with lead paint.

2.02 ACTIVITES PRIOR TO WORK

A. Lead-Based Paint Removal Work Shall Not Commence Until:
   1. Arrangements have been made for disposal of waste at an EPA a State approved landfill and the Forest Service has provided an EPA ID Number.
   2. Arrangements have been made for containing and/or disposal of wastewater resulting from showering and other abatement activities.
   3. Containment, decontamination enclosure system, and waste load-out area are appropriately constructed.
   4. All tools, equipment, PPE, and material waste receptacles are on-site.
   5. Suitable barricades and warning signs are established to restrict access to the work area to abatement associated personnel only.
   6. All other preparatory steps have been taken and applicable notices posted, and permits are obtained.
   7. A visitor and employee log-in log-out system is in place at the job site. All persons entering the site will be required to sign in and sign out.
   8. All notifications for lead related work have been completed in accordance with Federal, State, and local requirements.

2.03 CONTROL OF EMISSIONS AND DUST

A. The ground and any plants or shrubs in the area in which demolition is occuring shall be covered with plastic and weighted at all edges so as to prevent blowing. The covering shall be placed in a manner that traps debris and water. The covering shall be properly disposed of and not re-used.
   1. Exterior work shall not be conducted when winds are in excess of 20 mph.

B. Caution Signs
   1. At each separate work area, the Contractor performing demolition shall display a caution sign wherever the treatment process is reasonably expected to break or disturb any lead-containing substances.
   2. At least 3 days before demolition of structures, the Contractor shall post signs immediately outside all entrances and exits to the work area.
   3. The Contractors shall keep the signs posted until all work has been copleted and any remaining LBP is encapsulated.
   4. The Contractor shall insure that the sign meets the following description:
   5. The sign is at least 20" by 14" and states the date and place of the lead abatement project;

6. The sign includes the phrase, "Caution Lead Hazard, Keep Out" in bold lettering at least 2 inches high.

C. Warning Signs
   1. Warning signs in the work area shall confrom to 29 CFR 1926.62 (M) and Title 8 Code of California Regulation Section 1532.1 subsection (m).

2.04 ABATEMENT METHODS

A. The following methods shall not be used to remove paint that is, or may be, lead-based paint:
   1. open flame burning or torching,
   2. machine sanding or grinding without a high-efficiency particulate air (HEPA) local exhaust control,
   3. abrasive blasting or sandblasting without HEPA local exhaust control,
   4. heat guns operating above 1100 degrees Fahrenheit or charring the paint,
   5. dry sanding or dry scraping, except dry scraping in conjunction with heat guns or within 1.0 feet of electrical outlets, or when treating defective paint spots totaling no more than 2 sq. ft. (0.2 sq. m.) in any one interior room or space or totaling no more than 20 sq. ft. (2.0 sq. m.) on exterior surfaces,

B. paint stripping in a poorly ventilated space using a volatile stripper that is a hazardous substance in accordance with regulations of the Consumer Product Safety Commission at 16 CFR 1500.3, and/or a hazardous chemical in accordance with the Occupational Safety and Health Administration regulations at 29 CFR 1910.1200 or 1926.59, as applicable to the work.

2.05 CLEAN UP PROCEDURES

A. The work site shall be cleaned at end of each day's abatement activities. A separate, secured area within the project site shall be designated for storage of debris until it can be properly disposed of. Disposable supplies shall be replaced regularly and disposed of properly. Durable equipment, such as power and hand tools, generators, and vehicles shall be cleaned monthly or prior to removal from the site. All equipment shall be cleaned by HEPA vacuuming and high-phosphate washing.

B. Clean-Up Methods and Equipment
   1. Large debris from demolition (i.e. doors, windows, baseboards) shall be wrapped in plastic sheets at lest 6 mil thick, sealed with heavy duct tape, and stored in the designated area. This debris will be removed according to specifications outlined in Section 2.04 contract specifications.
   2. Small debris will be picked up, collected, and placed into a single plastic bag, at least 6 mil thick, or double bags of 4 mil thick plastic. The bags shall not be overloaded, shall be securely tied, and shall be stored in the designated area until disposal.
   3. Plastic sheeting shall be sprayed, picked up, and HEPA vacuumed prior to removal. The plastic sheeting shall be carefully folded from the corners and ends toward the middle and placed into a double 4 mil or single 6 mil plastic bag and sealed. Bags shall be stored in the designated area and disposed according to contract specifications.
   4. The Contractor shall removal all plastic sheeting for preclearance cleanup. Visible debris in the soil shall be removed.
   5. The Contractor shall await analysis results of baseline and post-hazard control soil samples. If post-hazard control soil levels are greater than or equal to 1000 ug/g, baseline soil samples will be analyzed to determine the additional work that will be needed. If paint chips originating from the work are identified in the soil, they shall be removed with a HEPA vacuum.
   6. The Contractor shall begin final cleanup no less than 24 hours (unless notified otherwise) after final clearance and all materials, equipment, debris, and plastic sheeting have been removed. All disposable supplies used during cleanup shall be disposed of according to Section 2.04 of these specifications.

2.06 POST ABATEMENT CLEAN UP AND INTERIOR CLEARANCE

A. Final clean-up prior to re-occupancy shall include wet wiping using a TSP solution and HEPA vacuuming all suspect dust and debris for final visual inspection and wipe dust sampling. All cleaning will be performed by the abatement contractor.

B. Visual inspections and clearance sampling (wipe sampling) will be performed by an independent third party to ensure that abatement areas are safe for re-occupancy. If the most stringent of EPA, or HUD or State of California clearance standards are not met, the Contractor must return to the project site and perform additional cleaning. This process will be repeated until the applicable clearance standards are achieved.

C. If clearance samples fail the Contractor is responsible for all subsequent consulting and testing costs until the clearance is achieved. The Contractor shall remedy the situation that caused the failure, i.e., filter outside dirty air, re-clean inside surfaces or other, and then the containment area shall be retested. Once the work areas have achieved clearance, the Contractor shall remove containment barriers.

2.07 POST ABATEMENT CLEAN UP AND EXTERIOR CLEARANCE

A. Exterior clearance shall include a visual examination of the ground, vegetation, and horizontal building surfaces near the exterior work surfaces to confirm that paint chips and other paint-related debris have been removed. Outdoor, hard surfaced living areas within the clearance area shall be visually examined to ensure these areas are free of visible dust and debris. The Contractor is responsible for exterior post abatement cleaning.

B. Visual examinations will be performed by an independent third party to ensure abatement areas are safe for occupancy.

C. If exterior cleanup is necessary, the Contractor is responsible for cleaning up the exterior areas. Clearance has not been achieved until such cleanup has been satisfactorily completed.

2.08 WASTE DISPOSAL

A. This section covers the disposal of waste generated during the demolition of structures containing lead-based paint.

B. The Contractor shall remove and dispose of in a legal manner all hazardous and nonhazardous waste generated. Waste (both hazardous and nonhazardous) shall be properly labeled, stored and secured while on site. Waste shall be removed from the project site for proper disposal within one week of project completion. At no time will waste be allowed to accumulate on-site for a period of more than 90 days.

C. The Contractor shall contact the regional EPA, State, and local authorities to determine current lead-based paint debris disposal requirements. The requirements of the Resource Conservation and Recovery Act (RCRA) shall be complied with as well as applicable state solid waste plan requirements. During demolition, the Contractor shall not leave debris in the yard or nearby property, incinerate debris, dump waste by a road or in an unauthorized dumpster, or introduce lead-contaminated water into storm drains or sanitary sewers.

D. Testing on lead-based paint abatement waste materials by use of the TCLP method will be completed by the Contractor and results supplied to the U.S. Forest Service.

E. Forest Service will obtain and provide an EPA number to use for disposal of all hazardous material produced as part of project scope.

F. The Contractor shall place lead-based paint chips, debris and lead dust in double (4-mil) or single (6-mil) polyethylene bags that are air-tight and puncture-resistant. Pieces of wood or other types of substrates that do not fit into plastic bags will be wrapped and labeled "DANGER, LEAD DUST".

G. The Contractor will place all disposable cleaning materials, such as sponges, mop heads, filters, disposable clothing, and brooms in double (4-mil) or single (6-mil) plastic bags and seal.

H. The Contractor shall clean surfaces and equipment and bag large debris. The Contractor shall then remove plastic sheeting and tape from covered surfaces. Prior to removig the plastic

sheeting, the Contractor shall lightly mist the sheeting in order to keep dust down and fold inward to form tight small bundles to bag for disposal. The Contractor shall place all plastic sheeting in double (4-mil) or single (6-mil) thick plastic bags and seal.

I. The contractor shall bag and seal vacuum bags and filters in double (4-mil) or single (6-mil) thick plastic bags.

J. The Contractor shall place all contaminated clothing or clothng covers used during demolition and cleanup in plastic bags for disposal.

K. The Contractor shall contain and properly dispose of all liquid waste, including lead-dust contaminated wash water.

L. The Contractor shall HEPA vacuum the exterior of all liquid waste containers prior to removing the waste containers from the work area and shall wet wipe the containers to ensure that there is no residual contamination. Containers should then be moved out of the work area into the designated storage area. The Contractor shall carefully place the containers into the truck or dumpster used for disposal.

M. The Contractor shall ensure that all waste is transported in covered vehicles to a landfill, or lined landfill.

N. If the Contractor subcontracts the removing of the lead-based paint waste, he shall insure that the company removing the waste material adequately covers all loads so as to assure that no dust or debris is released.

O. The Forest Service shall supply the contractor with an EPA ID Number.

P. If the Contractor is not a licensed hazardous waste transporter, a contract shall be entered into with a transporter licensed to haul hazardous waste.

2.09 NOTIFICATION

A. Contractor is responsible for providing written notification to Cal/OSHA twenty four (24) hours before conducting lead-related construction work involving trigger tasks identifiied in Title 8 Section 1532. Copies of the notification indicating the date and time submitted shll be provided to the Forest Service Contracting Officer

2.10 WORKER PROTECTION

2.11 WORKERS PERFORMING LEAD RELATED CONSTRUCTION ACTIVITIES MUST HAVE CURRENT STATE OF CALIFORNIA CERTIFICATION TO PERFORM SUCH WORK.

A. All workers must have baseline and post demolition blood lead level screenings determined by the whole blood lead method. The blood tests will also be required at the end of each 30 day period, if the employee is working on the job. This test must be performed before workers enter a lead contaminated work areas. A worker will be removed from the job if his blood lead level is 50 ug/dl or greater. The Contractor shall be responsible for medical surveillance and record keeping.

B. The Contractor will provide all workers, foremen, and superintendents with properly fitted half-faced respirators approved by NIOSH and OSHA. When respirators and disposable filters are employed, sufficient replacement filters will be provided by the Contractor for the workers..All workers inside the work area will wear the proper respirator for the lead dust level generated.

C. All workers performing lead-based paint abatement shall have properly fitted half-face respirators approved by NIOSH and OSHA and meet all minimum respiratory protection requirements as regulated by OSHA when performing tasks that are above the permissible exposure level. Workers will wear full body disposable suite with hoods and booties and appropriate goggles with a side shield. Disposal suits shall be used once, then properly discarded.

D. The abatement Contractor shall follow and enforce good personal hygiene practices during abatement activities. Upon leaving the work area, each worker will remove and dispose of work suit, wash and dry face and hands and vacuum clothes.

END OF SECTION

SECTION 073100
SYNTHETIC ROOF TILES

PART 1 - GENERAL

1.01 SECTION INCLUDES

A. Synthetic roof tiles of the following types:
   1. Shake tiles. (Cedar Shake)

B. Accessories.

1.02 RELATED REQUIREMENTS

A. Section 06 10 00 - Rough Carpentry.

B. Section 07 60 00 - Flashing and Sheet Metal.

1.03 REFERENCE STANDARDS

A. ASTM International (ASTM):
   1. ASTM D226 - Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing.
   2. ASTM D3161 - Standard Test Method for Wind-Resistance of Asphalt Shingles (Fan-Induced Method).
   3. ASTM E108 (UL 790) - Standard Test Methods for Fire Tests of Roof Coverings.
   4. ASTM D1970 - Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection.

B. International Code Council (ICC): ES Acceptance Criteria AC07 Section 4.9.

C. Underwriters Laboratories (UL):
   1. UL 790 - (Exterior Exposure), Standard Test Methods for Fire Tests of Roof Coverings.
   2. UL 997 - Wind Resistance of Prepared Roof Covering Materials.
   3. UL 2218 - Impact Resistance of Prepared Roof Covering Materials.

D. Manufacturer's recommendations and specifications.

1.04 SUBMITTALS

A. See Section 01 33 00 – Submittal Procedures, for submittal procedures.

B. Product Data:
   1. Manufacturer's data sheets on each product to be used.
   2. Preparation instructions and recommendations.
   3. Storage and handling requirements and recommendations.
   4. Typical installation methods.

C. Selection Samples: For each finish product specified, two sample roof tiles representing the manufacturer’s full range of available colors and patterns.

D. Verification Samples: For each finish product specified, two samples, representing actual product, color, and patterns.

E. Shop Drawings: Include details of materials, construction, and finish. Include relationship with adjacent construction.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with a minimum years documented experience.

B. Installer Qualifications: Company specializing in performing Work of this section with minimum two years documented experience with projects of similar scope and complexity.

C. Source Limitations: Provide each type of product from a single manufacturing source to ensure uniformity.

D. Mock-Up: Construct a mock-up with actual materials in sufficient time for Government's review and to not delay construction progress. Locate mock-up as acceptable to Government and provide temporary foundations and support.
   1. Intent of mock-up is to demonstrate quality of workmanship and visual appearance.
   2. If mock-up is not acceptable, rebuild mock-up until satisfactory results are achieved.
   3. Retain mock-up during construction as a standard for comparison with completed work.
   4. Do not alter or remove mock-up until work is completed or removal is authorized.

1.06 PRE-INSTALLATION CONFERENCE

A. Convene a conference approximately two weeks before scheduled commencement of the Work. Attendees shall include Government, Contractor and trades involved. Agenda shall include schedule, responsibilities, critical path items and approvals.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Store and handle in strict compliance with manufacturer's written instructions and recommendations.

B. Protect from damage due to weather, excessive temperature, and construction operations.

C. Store and dispose of solvent-based materials, and materials used with solvent based materials, in accordance with requirements of local authorities having jurisdiction.

1.08 PROJECT CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's recommended limits.

1.09 WARRANTY

A. Manufacturer's Warranty: Provide manufacturer's standard limited warranty:
   1. Manufacturer's 50-year warranty for tiles against material defects that cause leaks under normal weather and use conditions.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Acceptable Manufacturer: Brava Roof Tile or approved equal.

2.02 PERFORMANCE REQUIREMENTS

A. Roof system to consist of manufactured synthetic tiles attached to structural substrate to form weather tight roof envelope with no measurable water penetration.

B. Tiles shall be manufactured with variations in color and size, textured faces and edges, and sufficient thickness to provide a realistic installed appearance.

C. Method of attachments shall be designed to adequately resist wind uplift for roof configuration and project location.

D. Fire resistance: When installed over one layer of 30 lbs. felt that meets ASTM D226

E. Type II standard or a listed synthetic underlayment: Class C tested in accordance with ASTM E108/UL 790. See Code Evaluation Report for specific fire classified assemblies.

F. Fire resistance: When installed over one layer of GAF VersaShield Fire Resistant Underlayment or approved equal. Class A tested in accordance with ASTM E108/UL 790. Meet product Code Evaluation Report for specific fire classified assemblies.

G. Wind Uplift Resistance: When installed over minimum 15/32-inch-thick plywood. Tested in accordance with method TAS 125.

H. Hail Impact Resistance: Class 4 to withstand two drops of 2 inches diameter, 1.2 lbs. (0.54 kg) steel ball dropped from 20 feet tested in accordance with UL 2218.

I. Freeze-thaw resistance: No crazing, cracking, delamination of coating, or other deleterious surface changes after one month exposure with temperature cycled from

J. -40 to in 22 hours tested in accordance with International Code Council (ICC)- ES Acceptance Criteria AC07 Section 4.9.

K. Accelerated weathering: Little change after 2,500 hours exposure to ultraviolet (UV) radiation, elevated temperature, moisture, and thermal shock.

2.03 SYNTHETIC SHAKE TILES

A. Basis of Design: Brava Roof Tile, Cedar Shake as manufactured by Brava Roof Tile.

B. Lightweight, synthetic shake shingles with the appearance, color, texture, and thickness of natural wood shakes

C. Material: Engineered polymer formulated from post-industrial recycled plastic.

D. Installed weight: At 10-inch 10 inch exposure: 304 lbs. per 100 sq feet.

E. Profile: Rectangular shape with exposed to view upper surface and edges textured to
   1. resemble natural wood shake. Underside formed with reinforcing ribs for added strength and stability.

F. Size:
   1. Thickness: 3/8 to 1 inch (9.5 to 25 mm) at butt end.
   2. Length: 22 inches.
   3. Variable widths: 5, 7, and 12 inches.

G. Starter tile: 12 inches long by 12 inches wide.

H. Markings: Form tiles with markings on upper surface to indicate nailing locations and provide alignment guidelines for different exposure lengths.

I. Finish: Multi color comparable to natural wood shakes. Internal UV stabilizers to provide durable color stability.
   1. Color: Canyon Gray.
   2. Color: White (Premium color).

2.04 ACCESSORLES

A. Underlayment: Asphalt saturated and coated organic felt base sheet which meets or exceeds requirements of ASTM D226 Type II No. 30 non-perforated saturated asphalt felt or a listed synthetic underlayment which meets or exceeds requirements of ASTM D8257, in accordance with manufacturer's instructions.
   1. Approved synthetic membranes include, but are not limited to:
      a. Titanium UDL 50.
      b. Low-E ThermaSheet
      c. Eco Chief SOLARHIDE-SRW
      d. GAF VersaShield Resistant Roof Deck Protection

B. Waterproof Sheet Membrane: Cold applied, self-adhering waterproof membrane composed of polyethylene film coated one side with rubberized asphalt adhesive which meets or exceeds requirements of ASTM D1970, in accordance with manufacturer's instructions.
   1. Thickness: 40 mils 0.04 inch.
   2. Low temperature flexibility: Unaffected at minus 32 degrees Fahrenheit.
   3. Minimum tensile strength: 250 psi.
   4. Minimum elongation: 250 percent.
   5. Permeance: 0.05 perms maximum.
   6. Approved membranes include, but are not limited to:
      a. Titanium PSU 30.

C. Flashing: Fabricate from sheet to profiles and dimensions indicated on Drawings and approved shop drawings and in accordance with general requirements specified in Section 07 60 00 - Flashing and Sheet Metal- Flashing and Sheet Metal.
   1. Material: 16-ounce copper.
   2. Material: 26 gage 0.02 inch galvanized steel.

3. Base Flashings: To be used over or under the roof coverings and are turned up on the vertical surface.
    a. Base flashings should extend under the uppermost row of tile the full depth of the tile or at least 4 inches over the tile immediately below the metal. The vertical leg of the metal should be turned up a minimum of 4 inches and extend 4 inches on the tile as it is laid.
4. Linear Components: Form in longest possible lengths with 8 feet as minimum.
5. Counter Flashings: Extend 4 inches minimum up vertical surfaces
6. and 4 inches minimum under shingles.
7. Eave Flashings: Form bottom edge outward 1/4-inch 0.24 inch and hem to form drip.
8. Valley Flashings: 24 inches minimum width and extending 10 inches minimum from valley center line.

D. Standard Fasteners: 3/8-inch 0.37 inch diameter, corrosion resistant flat head ring shank roofing nails 1-3/4 inches long.
1. Material: Copper.
2. Material: Stainless steel.
3. Material: Hot-dipped galvanized.

E. Standard Fasteners: No. 10 DSVT T25 screws 2 inches long.
1. Material: Copper.
2. Material: Stainless steel.
3. Material: Hot-dipped galvanized.

F. High wind fasteners (90,100, 130 mph): 1/8-inch 0.12 inch diameter, No. 8 x 2-inch (2.01 inch) screws.
1. Material: Copper.
2. Material: Stainless steel.
3. Material: Hot-dipped galvanized.

G. High wind fasteners (90,100, 130 mph): No. 10 x 2-1/2-inch 2.52 inch screws.
1. Epoxy, polymer or ceramic coated.
2. Material: Stainless steel.
3. Material: Hot-dipped galvanized.

H. Snow Guards
1. Alpine Fusion-Guard Hybrid Pad-Style Snow Guard.
2. Snow Gem Half Carat™ Metal Snow Guard with Plate
3. TRA Snow & Sun - Snow Bracket Brava - APEX

PART 3 - EXECUTION

3.01 EXAMINATION

A. Do not begin installation until substrates have been properly constructed and prepared.

B. If substrate preparation is the responsibility of another installer, notify Government in writing of unsatisfactory preparation before proceeding.

3.02 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Coordinate installation with provision of gutters and downspouts as specified herein.

D. Verify roof is complete, rigid, braced, and deck members are securely fastened. Ensure proper ventilation has been provided for roof space. Do not proceed with roofing until deficiencies are addressed.

E. Verify roof deck is clean, dry, and ready to receive Brava synthetic Shake or approved equal.

3.03 INSTALLATION

A. Install products in accordance with manufacturer's instructions, approved submittals and in proper relationship with adjacent construction.

B. Underlayment Installation: Install self-adhered waterproof sheet membrane on the eaves. Cover the waterproof sheet membrane and the remaining portions of the roof as scheduled with the approved underlayment. Then install waterproof sheet membrane in valleys, along walls and around projections terminating on top of underlayment. Install underlayment in accordance with underlayment and tile manufacturers' recommendations.
   1. Stripping ply: Install full sheet of self-adhered waterproof sheet membrane in valleys, and minimum 18 inch width on gable ends, against walls, and around projections.
   2. In areas where January average daily temperature is 25 degrees Fahrenheit (minus 4 degrees
      a. C) or lower or where ice buildup is possible, install self-adhered waterproof sheet membrane from bottom edge extending two feet 24.02 inch above exterior wall line on eaves.
   3. Install waterproof sheet membrane over entire roof area.
      a. Apply waterproof sheet membrane at temperatures of 40 degrees Fahrenheit (4 degrees
         1) C) or higher.
      b. Adhere and attach as recommended by manufacturer of waterproof sheet membrane.
      c. Start underlayment installation at lower edge of roof. Install perpendicular to roof slope with minimum 4-inch 4.02 inch side laps and minimum 6- inch (5.98 inch) end laps.
      d. Extend underlayment minimum 4 inches up vertical wall intersections and at least 6 inches onto the roof.
      e. Do not leave underlayment membrane exposed in excess of time limit required by manufacturer. Do not puncture or tear underlayment.

C. Flashing Installation:
   1. Install overhanging drip edge on eaves and gable ends and metal flashings at valleys, ridges, hips, roof curbs, penetrations, and intersections with vertical surfaces in accordance with Section 07 60 00 - Flashing and Sheet Metal- Flashing and Sheet Metal.
   2. Weather lap joints 2 inches minimum and seal with sealant as specified in Section 07 91 13 - Compression Seals- Joint Fillers.
   3. Secure in place with clips, nails, and other fasteners.

3.04 TILE INSTALLATION- GENERAL

A. Install tiles in accordance with manufacturer's instructions and approved Shop Drawings.

B. Layout: Accurately layout tiles. Ensure that edges are parallel and perpendicular to roof eaves.

C. Confirm roof square before installation. Correct any out of square conditions and ensure that the eave line is straight.

D. Color: Verify that materials are the selected color. Some color variation is to be expected. If color variation exceeds original selection, immediately cease installation, and notify manufacturer.

E. Selection: Pull materials from multiple pallets to ensure proper color blending.

F. Roof Slope: Minimum roof slope recommended for the application of synthetic roof tile is
   1. 4:12. When installed on a 3:12 slope, a self-adhered waterproof membrane should be used on the entire slope.

3.05 TILE INSTALLATION - SHAKE

A. Spacing: Provide a minimum 3/16-inch 0.19 inch gap between all tiles to allow for expansion and contraction. A 3/8-inch (9.5) spacing is recommended.

B. Exposure: Maximum 10 inches. Minimum 4 inches.

C. Fastening: Attach each tile to wood deck with 2 approved fasteners.
   1. Place fasteners at locations indicated on tiles.
   2. Use approved nails or screws for the tile type being installed.
   3. Ensure full penetration but do not overdrive fasteners.
   4. Ensure min. 3/4-inch 0.75 inch fastener penetration or completely through the deck.
   5. Do not fasten at an angle.
   6. Ensure that the fastener head is flush with shingle surface.
   7. At valleys do not fasten tiles within 5 inches of valley center line.

D. Cutting: Layout work to avoid cutting shingles.
   1. At gables and vertical intersections, vary combination of shingle width and spacing of shingles to avoid cutting.
   2. If cutting is required, place shingle such that cut edge is not exposed. The use of solid field tiles is recommended be best aesthetics.
   3. Use circular saw or straight edge and utility knife if cuts are necessary.

E. Sidelap: Stagger tiles joints in one course minimum 1-1/2 inches from joints in course below.

F. Eaves: Install row of starter tiles at eaves as base layer. Overhang starter tile at eave edge approximately 1/2 inch and 1 inch past rake edge, or as required to allow water to drain.

G. Ridges and Hips: After field shingle installation is complete, install an 8-inch-wide 7.99 inch strip of self-adhering membrane over the center of the hip and any unvented sections of ridge.
   1. Install ridge or hip caps using the correct profile for correct fit (Low, Standard, Steep).
   2. Install Hip/Ridge Caps with two fasteners at the formed fastener locators penetrating
      a. ≥ 3/4 inches into the deck or completely through the deck.
   3. Match Hip/Ridge exposure to Field Shake exposure ≤ 10 inches.
   4. 3-inch 2.99 inch fasteners recommended for hip and ridge caps

H. Valleys: Install full 36-inch 35.98 inch sheet of self-adhered waterproof sheet membrane in valleys.
   1. Open Valley: A minimum 4-inch 4.02 inch opening at the top of the valley is recommended using Solid Shakes with a “W” style flashing and 1-inch center crimp. If Solid Shakes are not used with 1 1/2-inch (1.5 inch) crimps, a “Double W” flashing may be used to cover exposed structural ribbing on cut Field Shakes.
   2. Closed Valley: A “W” style flashing may be used with a 1 1/2-inch 1.5 inch center crimp. Leave a minimum 3/16-inch (0.19 inch) gap between the shakes at the valley for thermal expansion.
   3. At valleys do not fasten tiles within 5 inches of valley center line.

3.06 FIELD QUALITY CONTROL

A. Field Inspection: Coordinate field inspection in accordance with appropriate sections in
   1. Division 01.

B. Inspect tiles as they are installed. Do not install cracked, broken, twisted, curled, or otherwise damaged tiles.

C. As work progresses, exercise care not to scratch or mark installed tiles. Tiles damaged during installation shall be immediately removed and discarded.

D. After approximately 200 sq feet of tiles have been installed, inspect roof from ground. Verify proper layout and appearance. Repeat inspection throughout installation.

E. Visually inspect complete installation to ensure that it is weather tight.

3.07 3CLEANING AND PROTECTION

A. Clean products in accordance with the manufacturer's recommendations.

B. Protect installed products until completion of project.

C. Color match accessories for aesthetic purposes if desired.

D. Touch-up, repair or replace damaged products before Substantial Completion.

END OF SECTION

SECTION 081113
HOLLOW METAL DOORS AND FRAMES

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Non-fire-rated hollow metal doors and frames.

1.02 RELATED REQUIREMENTS

A. Section 087100 - Door Hardware.

1.03 REFERENCE STANDARDS

A. ANSI/SDI A250.3 - Test Procedure and Acceptance Criteria for Factory Applied Finish Coatings for Steel Doors and Frames; 2025.

B. ANSI/SDI A250.4 - Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames and Frame Anchors; 2024.

C. ANSI/SDI A250.8 - Specifications for Standard Steel Doors and Frames (SDI-100); 2023.

D. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames; 2025.

E. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2025a.

F. ASTM A1008/A1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable; 2025.

G. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2025.

H. BHMA A156.115 - Hardware Preparation in Steel Doors and Frames; 2016.

I. NAAMM HMMA 830 - Hardware Selection for Hollow Metal Doors and Frames; 2002.

J. NAAMM HMMA 831 - Hardware Locations for Hollow Metal Doors and Frames; 2024.

K. NAAMM HMMA 861 - Guide Specifications for Commercial Hollow Metal Doors and Frames; 2014.

L. SDI 117 - Manufacturing Tolerances for Standard Steel Doors and Frames; 2023.

1.04 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Materials and details of design and construction, hardware locations, reinforcement type and locations, anchorage and fastening methods, and finishes.

C. Shop Drawings: Details of each opening, showing elevations, glazing, frame profiles, and any indicated finish requirements.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Hollow Metal Doors and Frames:
   1. Ceco Door, an Assa Abloy Group company; _____: www.assaabloydss.com/#sle.
   2. Curries, an Assa Abloy Group company; _____: www.assaabloydss.com/#sle.
   3. Deansteel Manufacturing Company, Inc; Hollow Metal Doors - SP Series: www.deansteel.com/#sle.
   4. De La Fontaine; HC - Honeycomb Core: www.delafontaine.com/#sle.
   5. Dunbarton Corporation; Hollow Metal: www.dunbarton.com/#sle.
   6. Fleming Door Products, an Assa Abloy Group company; _____: www.assaabloydss.com/#sle.

7. Krieger Specialty Products; _____: www.kriegerproducts.com/#sle.
8. Mesker, dormakaba Group; FDJ Series Drywall Frames: www.meskeropeningsgroup.com/#sle.
9. Premier Steel Doors and Frames: www.trustpremier.com/#sle.
10. Republic Doors, an Allegion brand; ____: www.republicdoor.com/#sle.
11. Steelcraft, an Allegion brand; ____: www.allegion.com/#sle.
12. Taylor Entrance Systems; Cambridge: www.taylordoor.com/#sle.
13. Technical Glass Products; SteelBuilt Window & Door Systems: www.tgpamerica.com/#sle.
14. Titan Metal Products, Inc; Builders Series 20 - 90 Minute Doors: www.titanmetalproducts.com/#sle.
15. Or approved equal..

2.02 PERFORMANCE REQUIREMENTS

A. Requirements for Hollow Metal Doors and Frames:
1. Steel Sheet: Comply with one or more of the following requirements; galvannealed steel complying with ASTM A653/A653M, cold-rolled steel complying with ASTM A1008/A1008M, or hot-rolled pickled and oiled (HRPO) steel complying with ASTM A1011/A1011M, commercial steel (CS) Type B, for each.
2. Hardware Preparations, Selections and Locations: Comply with NAAMM HMMA 830 and NAAMM HMMA 831 or BHMA A156.115 and ANSI/SDI A250.8 (SDI-100) in accordance with specified requirements.

B. Combined Requirements: If a particular door and frame unit is indicated to comply with more than one type of requirement, comply with the specified requirements for each type; for instance, an exterior door that is also indicated as being sound-rated must comply with the requirements specified for exterior doors and for sound-rated doors; where two requirements conflict, comply with the most stringent.

2.03 HOLLOW METAL DOORS

A. Door Finish: Factory primed and field finished.

B. Exterior Doors: Thermally insulated.
1. Based on SDI Standards: ANSI/SDI A250.8 (SDI-100).
a. Level 2 - Heavy-duty.
b. Physical Performance Level B, 500,000 cycles; in accordance with ANSI/SDI A250.4.
c. Model 1 - Full Flush.
d. Door Face Metal Thickness: 16 gauge, 0.053 inch, minimum.
2. Door Thickness: 1-3/4 inches, nominal.
3. Door Face Sheets: Flush.

2.04 HOLLOW METAL FRAMES

A. Comply with standards and/or custom guidelines as indicated for corresponding door in accordance with applicable door frame requirements.

B. Frame Finish: Factory primed and field finished.

C. Exterior Door Frames: Knock-down type.
1. Frame Metal Thickness: 18 gauge, 0.042 inch, minimum.
2. Weatherstripping: Separate, see Section 087100.

2.05 FINISHES

A. Primer: Rust-inhibiting, complying with ANSI/SDI A250.10, door manufacturer's standard.

B. Factory Finish: Complying with ANSI/SDI A250.3, manufacturer's standard coating.
1. Color: As indicated on drawings.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify that opening sizes and tolerances are acceptable.

C. Verify that finished walls are in plane to ensure proper door alignment.

3.02 INSTALLATION

A. Install doors and frames in accordance with manufacturer's instructions and related requirements of specified door and frame standards or custom guidelines indicated.

B. Coordinate frame anchor placement with wall construction.

C. Install door hardware as specified in Section 087100.

D. Touch up damaged factory finishes.

3.03 TOLERANCES

A. Clearances Between Door and Frame: Comply with related requirements of specified frame standards or custom guidelines indicated in accordance with SDI 117 or NAAMM HMMA 861.

B. Maximum Diagonal Distortion: 1/16 inch measured with straight edge, corner to corner.

3.04 ADJUSTING

A. Adjust for smooth and balanced door movement.

END OF SECTION

SECTION 081416
FLUSH WOOD DOORS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Flush wood doors; flush and flush glazed configuration;.

1.02 RELATED REQUIREMENTS

A. Section 087100 - Door Hardware.

1.03 REFERENCE STANDARDS

A. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

B. AWMAC/WI (NAAWS) - North American Architectural Woodwork Standards; 2021, with Errata.

1.04 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Indicate door core materials and construction; veneer species, type and characteristics.

C. Shop Drawings: Show doors and frames, elevations, sizes, types, swings, undercuts, beveling, blocking for hardware, factory machining, factory finishing, cutouts for glazing and other details.

D. Warranty, executed in Owner's name.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Package, deliver and store doors in accordance with specified quality standard.

B. Accept doors on site in manufacturer's packaging, and inspect for damage.

C. Protect doors with resilient packaging sealed with heat shrunk plastic; do not store in damp or wet areas or areas where sunlight might bleach veneer; seal top and bottom edges with tinted sealer if stored more than one week, and break seal on site to permit ventilation.

1.06 WARRANTY

A. Manufacturer Warranty: Provide manufacturer's warranty on interior doors for the life of the installation. Complete forms in Owner's name and register with manufacturer.
   1. Include coverage for delamination of veneer, warping beyond specified installation tolerances, defective materials, and telegraphing core construction.

PART 2 PRODUCTS

2.01 DOORS

A. Doors: See drawings for locations and additional requirements.

B. Exterior Doors: Flush solid core construction and water repellent treated.
   1. Thickness: 1-3/4 inches, unless otherwise indicated.

2.02 DOOR AND PANEL CORES

A. Non-Rated Solid Core and 20 Minute Rated Doors:

2.03 DOOR FACINGS

A. Veneer Facing for Opaque Finish: Closed grain hardwood veneer, in compliance with indicated quality standard.

2.04 DOOR CONSTRUCTION

A. Fabricate doors in accordance with door quality standard specified.

B. Cores Constructed with stiles and rails:

C. Glazed Openings: Non-removable stops on non-secure side; sizes and configurations as indicated on drawings.

D. Factory machine doors for hardware other than surface-mounted hardware, in accordance with hardware requirements and dimensions.

E. Factory fit doors for frame opening dimensions identified on shop drawings, with edge clearances in accordance with specified quality standard.
   1. Exception: Doors to be field finished.

F. Provide edge clearances in accordance with the quality standard specified.

2.05 FINISHES - WOOD VENEER DOORS

A. Finish work in accordance with AWI/AWMAC/WI (AWS) or AWMAC/WI (NAAWS), Section 5 - Finishing for grade specified and as follows:
   1. Opaque:
      a. System - 4, Latex Acrylic, Water-based.
      b. Color: As selected by Government.
      c. Sheen: Semigloss.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify that opening sizes and tolerances are acceptable.

C. Do not install doors in frame openings that are not plumb or are out-of-tolerance for size or alignment.

3.02 INSTALLATION

A. Install doors in accordance with manufacturer's instructions and specified quality standard.

B. Field-Finished Doors: Trimming to fit is acceptable.
   1. Adjust width of non-rated doors by cutting equally on both jamb edges.
   2. Trim maximum of 3/4 inch off bottom edges.

C. Use machine tools to cut or drill for hardware.

D. Coordinate installation of doors with installation of frames and hardware.

3.03 ADJUSTING

A. Adjust doors for smooth and balanced door movement.

END OF SECTION

SECTION 087100
DOOR HARDWARE

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Hardware for wood and hollow metal doors.

B. Thresholds.

C. Weatherstripping and gasketing.

1.02 REFERENCE STANDARDS

A. ADA Standards - 2010 ADA Standards for Accessible Design; 2010.

B. BHMA A156.1 - Standard for Butts and Hinges; 2025.

C. BHMA A156.2 - Bored and Preassembled Locks and Latches; 2022.

D. BHMA A156.18 - Standard for Materials and Finishes; 2020.

E. BHMA A156.21 - Thresholds; 2025.

F. BHMA A156.22 - Standard for Gasketing; 2021.

G. BHMA A156.36 - Auxiliary Locks; 2020.

1.03 ADMINISTRATIVE REQUIREMENTS

A. Coordinate the manufacture, fabrication, and installation of products that door hardware is installed on.

B. Sequence installation to ensure utility connections are achieved in an orderly and expeditious manner.

1.04 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Manufacturer's catalog literature for each type of hardware, marked to clearly show products to be furnished for this project, and includes construction details, material descriptions, finishes, and dimensions and profiles of individual components.

C. Shop Drawings - Door Hardware Schedule: Submit detailed listing that includes each item of hardware to be installed on each door. Use door numbering scheme as included in Contract Documents.
   1. Provide complete description for each door listed.
   2. Provide manufacturer name, product names, and catalog numbers; include functions, types, styles, sizes and finishes of each item.

D. Manufacturer's Installation Instructions: Indicate special procedures and perimeter conditions requiring special attention.

E. Maintenance Data: Include data on operating hardware, lubrication requirements, and inspection procedures related to preventative maintenance.

1.05 WARRANTY

A. Manufacturer's Warranty: Provide warranty against defects in material and workmanship for period indicated. Complete forms in Owner's name and register with manufacturer.
   1. Locksets and Cylinders: Three years, minimum.
   2. Other Hardware: Two years, minimum.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

A. Provide specified door hardware as required to make doors fully functional, compliant with applicable codes, and secure to extent indicated.

B. Provide individual items of single type, of same model, and by same manufacturer.

2.02 HINGES

A. Manufacturers:
   1. Schlage or equal.

B. Hinges: Comply with BHMA A156.1, Grade 1.
   1. Provide hinges on every swinging door.
   2. Provide following quantity of butt hinges for each door:
      a. Doors From 60 inches High up to 90 inches High: Three hinges.

2.03 CYLINDRICAL LOCKS

A. Manufacturers:
   1. Schlage, an Allegion brand: www.allegion.com/us/#sle.
   2. Or Approved Equal

B. Cylindrical Locks (Bored): Comply with BHMA A156.2, Grade 1, 4000 Series.
   1. Bored Hole: 2-1/8 inch diameter.
   2. Latchbolt Throw: 1/2 inch, minimum.
   3. Backset: 2-3/4 inch unless otherwise indicated.
   4. Strikes: Provide manufacturer's standard strike for each latchset or lockset with strike box and curved lip extending to protect frame in compliance with indicated requirements.
      a. Finish: To match lock or latch.

2.04 AUXILIARY LOCKS (DEADBOLTS)

A. Auxiliary Locks (Deadbolts): Comply with BHMA A156.36, Grade 1.
   1. Bolt Throw: 1 inch minimum, with latch made of hardened steel.

2.05 THRESHOLDS

A. Manufacturers:
   1. Pemko; an Assa Abloy Group company: www.assaabloydss.com/#sle.
   2. or approved equal. .

B. Thresholds: Comply with BHMA A156.21.
   1. Provide threshold at each exterior door, unless otherwise indicated.
   2. Type: Flat surface.
   3. Material: Aluminum.
   4. Threshold Surface: Fluted horizontal grooves across full width.
   5. Field cut threshold to profile of frame and width of door sill for tight fit.
   6. Provide non-corroding fasteners at exterior locations.

2.06 WEATHERSTRIPPING AND GASKETING

A. Manufacturers:
   1. Pemko; an Assa Abloy Group company: www.assaabloydss.com/#sle.
   2. Or approved equal.

B. Weatherstripping and Gasketing: Comply with BHMA A156.22.
   1. Head and Jamb Type: Self-adhesive.
   2. Door Sweep Type: Screw on.
      a. Material: Aluminum, with rubber weatherstripping.

2.07 FINISHES

A. Finishes: Provide door hardware of same finish, unless otherwise indicated.
   1. Finish: 625; bright chromium plated over nickel, with brass or bronze base material (former US equivalent US26); BHMA A156.18.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that doors and frames are ready to receive this work; labeled, fire-rated doors and frames are properly installed, and dimensions are as indicated on shop drawings.

3.02 INSTALLATION

A. Install hardware in accordance with manufacturer's instructions and applicable codes.

B. Use templates provided by hardware item manufacturer.

C. Door Hardware Mounting Heights: Distance from finished floor to center line of hardware item. As indicated in following list; unless noted otherwise in Door Hardware Schedule or on drawings.

1. Mounting heights in compliance with ADA Standards:

D. Set exterior door thresholds with full-width bead of elastomeric sealant at each point of contact with floor providing a continuous weather seal; anchor thresholds with stainless steel countersunk screws.

3.03 ADJUSTING

A. Adjust hardware for smooth operation.

B. Adjust gasketing for complete, continuous seal; replace if unable to make complete seal.

END OF SECTION

SECTION 099000
PAINTING AND COATING - COMMERCIAL FACILITY GUIDE SPECIFICATION - SHERWIN-WILLIAMS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Surface preparation.

B. Interior painting and coating systems.

C. Exterior painting and coating systems.

D. Scope:
   1. Finish surfaces exposed to view, unless fully factory-finished and unless otherwise indicated.
      a. Exterior:
         1) Wood: Decks, exterior including pressure treated lumber, non-vehicular floors and platforms.
         2) Wood: Siding, trim, shutters, sashes, and hardboard-bare/primed.
      b. Interior:
         1) Wood: Walls, ceilings, doors, and trim.

1.02 REFERENCE STANDARDS

A. 40 CFR 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings; U.S. Environmental Protection Agency; Current Edition.

1.03 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Provide complete list of products to be used, with the following information for each:
   1. Product characteristics.
   2. Surface preparation instructions and recommendations.
   3. Primer requirements and finish specification.
   4. Storage and handling requirements and recommendations.
   5. Application methods.
   6. Clean-up information.

C. Maintenance Data: Submit coating maintenance manual including finish schedule showing where each product/color/finish was used, product technical data sheets, safety data sheets (SDS), care and cleaning instructions, touch-up procedures, repair of painted and finished surfaces, and color samples of each color and finish used.

1.04 QUALITY ASSURANCE

A. Applicator Qualifications: Company specializing in performing the type of work specified with minimum 3 years experience and approved by manufacturer.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Deliver products to site in sealed and labeled containers; inspect to verify acceptability.

B. Container Label: Include manufacturer's name, type of paint, product name, product code, color designation, VOC content, batch date, environmental handling, surface preparation, application, and use instructions.

C. Paint Materials: Store at a minimum of 45 degrees F and a maximum of 90 degrees F, in ventilated area, and as required by manufacturer's instructions.

1.06 FIELD CONDITIONS

A. Do not apply materials when environmental conditions are outside the ranges required by manufacturer.

B. Follow manufacturer's recommended procedures for producing the best results, including testing of substrates, moisture in substrates, and humidity and temperature limitations.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Basis of Design Products: Subject to compliance with requirements, provide Sherwin-Williams Company (The) products indicated; www.sherwin-williams.com/#sle.

B. Comparable Products: Products of approved manufacturers will be considered :
   1. Other Acceptable Manufacturers:
      a. Benjamin Moore & Co: www.benjaminmoore.com.

2.02 PAINTINGS AND COATINGS

A. General:
   1. Provide factory-mixed coatings unless otherwise indicated.
   2. Do not reduce, thin, or dilute coatings or add materials to coatings unless specifically indicated in manufacturer's instructions.

B. Volatile Organic Compound (VOC) Content:
   1. Provide paints and finishes that comply with the most stringent requirements specified in the following:
      a. 40 CFR 59, Subpart D--National Volatile Organic Compound Emission Standards for Architectural Coatings.

C. Accessory Materials: Provide primers, sealers, cleaning agents, cleaning cloths, sanding materials, and clean-up materials as required for final completion of painted surfaces.

2.03 PAINT SYSTEMS - EXTERIOR

A. Wood: Decks, exterior including pressure treated lumber, non-vehicular floors and platforms.
   1. Acrylic Floor Systems, Water Based:
      a. Floor Finish:
         1) 1st and 2nd Coat: Sherwin-Williams Porch and Floor Enamel, A32-200 Series: www.sherwin-williams.com/#sle.

B. Wood: Siding, trim, shutters, sashes, and hardboard-bare/primed.
   1. Latex Systems:
      a. Satin Finish:
         1) 1st Coat: Sherwin-Williams Latex Wood Primer, B42W8041: www.sherwin-williams.com/#sle or Benjamin Moore Primer
         2) 2nd and 3rd Coat: Sherwin-Williams A-100 Exterior Latex Satin, A82 Series: www.sherwin-williams.com/#sle or Benjamin Moore Element Guard Exterior Soft Gloss Finish 765

2.04 PAINT SYSTEMS - INTERIOR

A. Wood: Walls, ceilings, doors, and trim.
   1. Latex Systems:
      a. Eg-Shel/Satin Finish:
         1) 1st Coat: Sherwin-Williams Premium Wall and Wood Primer, B28W8111: www.sherwin-williams.com/#sle.
         2) 2nd and 3rd Coat: Sherwin-Williams ProClassic Waterborne Acrylic Satin, B20 Series: www.sherwin-williams.com/#sle.
   2. Stain and Varnish System:
      a. Satin Finish:
         1) 1st Coat: Sherwin-Williams Minwax Performance Series Tintable Wood Stain 250 VOC: www.sherwin-williams.com/#sle.
         2) 2nd and 3rd Coat: Sherwin-Williams Minwax Waterbased Oil-Modified Polyurethane: www.sherwin-williams.com/#sle.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that surfaces are ready to receive work as instructed by the product manufacturer.

3.02 PREPARATION

A. Clean surfaces thoroughly and correct defects prior to application.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Wood: Remove dust, grit, and foreign matter. Scrape, sand, and spot prime knots and pitch streaks. Fill nail holes and imperfections with wood filler and sand smooth.

3.03 APPLICATION

A. Remove unfinished louvers, grilles, covers, and access panels on mechanical and electrical components and paint separately.

B. Apply products in accordance with manufacturer's written instructions.

C. Apply coatings at spread rate required to achieve manufacturer's recommended dry film thickness.

D. Regardless of number of coats specified, apply additional coats until complete hide is achieved.

3.04 PRIMING

A. Apply primer to all surfaces unless specifically not required by coating manufacturer. Apply in accordance with coating manufacturer's instructions.

B. Primers specified in painting schedules may be omitted on items that are factory primed or factory finished if acceptable to top coat manufacturers.

3.05 CLEANING

A. Collect waste material that could constitute a fire hazard, place in closed metal containers, and remove daily from site.

B. Clean surfaces immediately of overspray, splatter, and excess material.

C. After coating has cured, clean and replace finish hardware, fixtures, and fittings previously removed.

3.06 PROTECTION

A. Protect finished coatings from damage until completion of project.

B. Touch-up damaged finishes after Substantial Completion.

END OF SECTION

SECTION 123530
RESIDENTIAL CASEWORK

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Kitchen cabinets.

B. Kitchen countertops.

1.02 SUBMITTALS

A. See Section 013000 - Administrative Requirements, for submittal procedures.

B. Product Data: Provide component dimensions, configurations, construction details, and joint details.

C. Shop Drawings: Indicate casework locations, elevations, clearances required, rough-in and anchor placement dimensions and tolerances.

D. Cabinet and Countertop Finish Sample: Submit two samples of each type of finish, 2 inches by 3 inches in size, illustrating color, texture, gloss, and wood species.

E. Warranty Documentation: Submit manufacturer warranty and ensure that forms have been completed in Owner's name and registered with manufacturer.

PART 2 PRODUCTS

2.01 CABINETS

A. Cabinet Box: Framed construction.
   1. Side Panels: Plywood.
      a. Exposed Side Panel Finish: Wood veneer, coordinate with cabinet door and drawer color/finish.
   2. Back Panel: Plywood.
   3. Bottom (and Top) Panel: Plywood.
   4. Face Frame: Solid wood.
   5. Interior Cabinet Finish: Natural plywood (unfinished).
      a. Color: To be coordinated with Government.
   6. Exposed Panel Edges: Finish with manufacturer's standard edge banding, color coordinated with other exposed finishes.

B. Cabinet Door/Drawer Configuration: Full overlay.

C. Cabinet Doors:
   1. Solid wood, painted finish.
   2. Paint Color: To Be Coordinated with Government.

D. Drawers:
   1. Solid wood sides with dovetail joints, plywood bottom panel.
   2. Drawer Front: To match cabinet doors in style, material, and finish.

E. Shelves: Manufacturer's standard adjustable shelves and shelf supports.

F. Cabinet Hardware: As selected from manufacturer's standard types, styles and finishes.

G. Kitchen Countertop: Post formed plastic laminate over particle board, coved to back splash. Color and finish to be coordinated with Government based of off manufacturer provided standard colors.

2.02 FABRICATION

A. Shop assemble casework for delivery to site in units easily handled and to permit passage through building openings.

B. Fabricate corners and joints without gaps.

C. Fabricate each unit to be rigid and not dependent on adjacent units for rigidity.

D. Provide cutouts for plumbing fixtures and appliances. Prime paint contact surfaces of cut edges.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install casework, components and accessories in accordance with manufacturer's instructions.

B. Use anchoring devices to suit conditions and substrate materials encountered.

C. Set casework items plumb and square, securely anchored to building structure.

D. Carefully scribe casework abutting other components, with maximum gaps of 1/8 inch.

END OF SECTION

SECTION 221005
PLUMBING PIPING

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Domestic water piping, above grade.

B. Propane gas piping, above grade.

1.02 REFERENCE STANDARDS

A. ASME B16.3 - Malleable Iron Threaded Fittings: Classes 150 and 300; 2021.

B. ASME B31.1 - Power Piping; 2024.

C. ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless; 2024.

D. ASTM A234/A234M - Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service; 2025.

E. ASTM F876 - Standard Specification for Crosslinked Polyethylene (PEX) Tubing; 2026.

F. ASTM F877 - Standard Specification for Crosslinked Polyethylene (PEX) Hot- and Cold-Water Distribution Systems; 2025.

G. NFPA 58 - Liquefied Petroleum Gas Code; 2024, with Amendment.

H. NSF 61 - Drinking Water System Components - Health Effects; 2024.

I. NSF 372 - Drinking Water System Components - Lead Content; 2024.

1.03 SUBMITTALS

A. See Section 013000 - Administrative Requirements for submittal procedures.

B. Product Data: Provide data on pipe materials, pipe fittings, valves, and accessories. Provide manufacturers catalog information. Indicate valve data and ratings.

C. Project Record Documents: Record actual locations of valves.

1.04 QUALITY ASSURANCE

A. Perform work in accordance with applicable codes.

B. Valves: Manufacturer's name and pressure rating marked on valve body.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

A. Potable Water Supply Systems: Provide piping, pipe fittings, solder, and flux (if used), complying with NSF 61 and NSF 372; label pipe and fittings.

2.02 DOMESTIC WATER PIPING, ABOVE GRADE

A. Cross-Linked Polyethylene (PEX) Pipe: ASTM F876 or ASTM F877.
   1. PEX Type, Fittings, and Joints:
      a. PEX-a, Engle method or PEX-b, Silane method; minimal fittings; use of continuous runs of piping prioritized.

2.03 PROPANE GAS PIPING, ABOVE GRADE

A. Steel Pipe: ASTM A53/A53M, Grade B, Type F, Schedule 40 black.
   1. Fittings: ASME B16.3, malleable iron, or ASTM A234/A234M, wrought steel welding type.
   2. Joints: NFPA 58, threaded or welded to comply with ASME B31.1.

PART 3 EXECUTION

3.01 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Install piping to maintain headroom, conserve space, and not interfere with use of space.

C. Group piping whenever practical at common elevations.

3.02 APPLICATION

A. Install unions downstream of valves and at equipment or apparatus connections.

B. Install ball valves for shut-off and to isolate equipment, part of systems, or vertical risers.

3.03 TOLERANCES

A. Drainage Piping: Establish invert elevations within 1/2 inch vertically of location indicated and slope to drain at minimum of 1/4 inch per foot slope.

3.04 FIELD TESTS AND INSPECTIONS

A. Verify and inspect systems according to requirements by the Authority Having Jurisdiction. In the absence of specific test and inspection procedures proceed as indicated below.

B. Domestic Water Systems:
   1. Perform hydrostatic testing for leakage prior to system disinfection.
   2. Test Preparation: Close each fixture valve or disconnect and cap each connected fixture.
   3. General:
      a. Fill the system with water and raise static head to 10 psi above service pressure. Minimum static head of 50 to 150 psi. As an exception, certain codes allow a maximum static pressure of 80 psi.

C. Test Results: Document and certify successful results, otherwise repair, document, and retest.

END OF SECTION

SECTION 263100
PHOTOVOLTAIC COLLECTORS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Photovoltaic system requirements.

B. Photovoltaic modules.

C. Photovoltaic module mounting system.

D. Photovoltaic combiner boxes.

E. Photovoltaic inverters.

F. Monitoring system.

1.02 REFERENCE STANDARDS

A. ASCE 7 - Minimum Design Loads and Associated Criteria for Buildings and Other Structures; Most Recent Edition Cited by Referring Code or Reference Standard.

B. IEC 61215-1 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1: Test Requirements; 2021, with Corrigendum.

C. IEC 61215-1-1 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1-1: Special Requirements for Testing of Crystalline Silicon Photovoltaic (PV) Modules; 2021.

D. IEC 61215-1-2 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1-2: Special Requirements for Testing of Thin-Film Cadmium Telluride (CdTe) Based Photovoltaic (PV) Modules; 2021, with Amendment (2022).

E. IEC 61215-1-3 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1-3: Special Requirements for Testing of Thin-Film Amorphous Silicon Based Photovoltaic (PV) Modules; 2021, with Amendment (2022).

F. IEC 61215-1-4 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 1-4: Special Requirements for Testing of Thin-Film Cu(In,GA)(S,Se)2 Based Photovoltaic (PV) Modules; 2021, with Amendment (2022).

G. IEC 61215-2 - Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval - Part 2: Test Procedures; 2021.

H. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2023.

I. NECA 412 - Standard for Installing and Maintaining Photovoltaic (PV) Power Systems; 2025.

J. NEMA EN 10250 - Enclosures for Electrical Equipment (1000 Volts Maximum); 2024.

K. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

L. UL 1699B - Outline of Investigation for Photovoltaic (PV) DC Arc-Fault Circuit Protection; Current Edition; Current Edition, Including All Revisions.

M. UL 1703 - Flat Plate Photovoltaic Modules and Panels; Current Edition, Including All Revisions.

N. UL 1741 - Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources; Current Edition, Including All Revisions.

1.03 SUBMITTALS

A. See Section 013000 - Administrative Requirements, for submittal procedures.

B. Design Documents: Prepare and submit all information required for plan review and permitting by authorities having jurisdiction, including but not limited to floor plans, riser diagrams, details, and description of operation.

C. Product Data: Provide manufacturer's standard catalog pages and data sheets for each product. Include ratings, configurations, standard wiring diagrams, outline and support point dimensions, finishes, weights, service condition requirements, and installed features.

D. Shop Drawings: Include dimensioned plan views and sections indicating locations of system components, required clearances, attachment locations and details, and proposed size, type, and routing of conduits and cables. Include system interconnection schematic diagrams showing all factory and field connections.

E. Design Data:
   1. Include structural calculations, certified by structural engineer, for equipment and mounting system.
   2. Include electrical calculations for array and associated equipment.

F. Certify that products of this section meet or exceed specified requirements.

G. Operation and Maintenance Data: Include detailed information on system operation, equipment programming and setup, replacement parts, and recommended maintenance procedures and intervals.

H. Warranty: Submit sample of manufacturer's warranty and documentation of final executed warranty completed in Owner's name and registered with manufacturer.

1.04 QUALITY ASSURANCE

A. Comply with NFPA 70.

B. Structural Designer Qualifications: Registered structural engineer licensed in the State in which the Project is located.

C. Electrical Designer Qualifications: Registered electrical engineer licensed in the State in which the Project is located experienced in the design of photovoltaic systems.

D. Installer Qualifications: Company specializing in performing the work of this section with minimum five years documented experience with photovoltaic systems of similar size, type, and complexity.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging, keep dry and protect from damage until ready for installation.

1.06 WARRANTY

A. Specified warranties indicate minimum requirements. Provide additional warranties or extended warranty periods where required to qualify for rebate and incentive programs.

B. Photovoltaic Modules:

C. Photovoltaic Combiner Boxes: Provide minimum five year manufacturer warranty covering repair or replacement due to defective materials or workmanship.

D. Photovoltaic Inverters: Provide minimum five year manufacturer warranty covering repair or replacement due to defective materials or workmanship.

E. Charge Controllers: Provide minimum five year manufacturer warranty covering repair or replacement due to defective materials or workmanship.

1.07 EXTRA MATERIALS

A. Provide (2) extra solar panels of the same specification and type installed to the Government prior to completion of the project.

PART 2 PRODUCTS

2.01 PHOTOVOLTAIC SYSTEM REQUIREMENTS

A. Design and provide complete photovoltaic system consisting of photovoltaic modules and associated balance of system components for a complete facility electrical system.

B. System Description:

1. Photovoltaic array is ground-mounted in location coordinated with Government.
2. System does not include interconnection with utility grid (off-grid system).
3. System includes battery storage system.
4. System includes monitoring system.

C. Size:
   1. Array: Array size is not critical.

D. Provide photovoltaic system and associated components suitable for wind loads, snow loads, seismic loads, and other structural design considerations of the installed location.
   1. Comply with ASCE 7.

E. Provide photovoltaic system and associated components suitable for continuous operation under the service conditions at the installed location.

F. Provide products listed, classified, and labeled as suitable for the purpose intended.

G. Unless specifically indicated to be excluded, provide all required equipment, conduit, boxes, wiring, connectors, hardware, supports, accessories, software, system programming, etc. as necessary for a complete operating system.

H. DC Arc Fault Circuit Protection: Provide DC photovoltaic arc-fault protection devices listed as complying with UL 1699B as required for compliance with NFPA 70.

I. Arrange equipment to provide minimum clearances in accordance with manufacturer's instructions and NFPA 70.

J. Arrange array to minimize shading during peak production periods.

2.02 PHOTOVOLTAIC MODULES

A. Acceptable Module Types: Either crystalline silicon or thin film modules complying with specified requirements will be considered for this project.

B. General Requirements:
   1. Photovoltaic Modules: Factory assembled; consisting of photovoltaic cells, frame, junction box, cables for series connection, and bypass diodes for shade tolerance; rated for 600 V DC; complying with IEC 61215-1 and IEC 61215-2 and listed as complying with UL 1703.
   2. Crystalline Silicon Photovoltaic Modules: Comply with IEC 61215-1-1.
   3. Thin Film Photovoltaic Modules: Comply with IEC 61215-1-2, IEC 61215-1-3, or IEC 61215-1-4 as applicable.
   4. Frame: Anodized aluminum.
   5. Factory-Installed Junction Box: Weatherproof, with factory-installed terminals and bypass diodes.
   6. Factory-Installed Cables: Type USE-2 or listed photovoltaic (PV) wire with polarized locking connectors.
   7. Unless otherwise indicated, specified module performance characteristics are rated under Standard Test Conditions (STC).

2.03 BALANCE OF SYSTEM COMPONENTS

A. Photovoltaic Module Mounting System:
   1. Provide complete mounting system compatible with modules to be installed and suitable to properly install them in the location indicated, including all necessary hardware and accessories.
   2. Support Structure and Associated Hardware Materials: Use aluminum, galvanized steel, or stainless steel.
   3. Ground-Mounted Arrays:
      a. Module Tilt Angle: As required to provide maximum energy production for installed location.
      b. Foundation Type: As required for soil conditions at installed location.

B. Photovoltaic Combiner Boxes:

1. Provide combiner box(es) for termination of strings as indicated or as required for the array configuration installed.
2. Combiner Boxes: Rated for 600 V DC; current ratings suitable for connected strings; equipped with fuseholders; listed as complying with UL 1741.
3. Fuseholders: Touch-safe; suitable to accept fuses indicated.
4. Number of Input Circuits: As indicated or as required for termination of strings, with minimum of 25 percent spare capacity for future expansion.
5. Enclosure: NEMA EN 10250, Type 3R, unless otherwise indicated.

C. Photovoltaic Inverters:
1. Provide inverter(s) as indicated or as required for connection of the photovoltaic array DC system to the AC system indicated.
2. Inverters: Suitable for the requirements of the connected array; output configuration compatible with connected system; listed as complying with UL 1741; furnished with the following features:
   a. Maximum power point tracking (MPPT).
   b. LCD display.
   c. Integral AC disconnect.
   d. Integral DC disconnect.
   e. Integral DC ground fault detection and interruption (GFDI).
3. Total Harmonic Distortion: Less than five percent.
4. Enclosure Environment Type per NEMA EN 10250: Unless otherwise indicated, as specified for the following installation locations:

D. Monitoring System:
1. Provide a system to monitor photovoltaic system performance including all sensors, dataloggers, connections, software, equipment and accessories necessary for a complete operating system.
2. System communications interfaces to be wired or wireless, with compatible interconnected components.
   a. Provide suitable raceway, minimum 3/4 inch trade size, for all required wired connections.
3. System to monitor and record, in 15 minute intervals:
   a. Inverter status.
   b. Instantaneous power (kW).
   c. Cumulative energy production (kWh).
4. System real-time and historical data to be accessible from the following locations:
   a. Personal computer(s), via internet connection.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that field measurements are as indicated.

B. Verify that ratings and configurations of system components are consistent with the indicated requirements.

C. Verify that mounting surfaces are ready to receive system components.

D. Verify that conditions are satisfactory for installation prior to starting work.

3.02 INSTALLATION

A. Perform work in accordance with NECA 1 (general workmanship).

B. Install products in accordance with manufacturer's instructions.

C. Provide required support and attachment in accordance with Section 260529.

D. Mount equipment such that the highest position of any operating handle for circuit breakers or switches does not exceed 79 inches above the floor, ground, or working platform.

E. Provide cast-in-place concrete foundations for ground-mounted arrays as required.

F. Circuiting Requirements. in Addition to Requirements of Section 260519:
   1. Wiring Methods:
      a. Unless otherwise indicated, use exposed module factory-installed cables (not routed inside building) for module interconnections.
      b. Secure exposed cables in accordance with NFPA 70. Where possible, conceal behind array.
   2. Photovoltaic DC System Conductor Color Code:
   3. Maintain separation of photovoltaic and non-photovoltaic circuits in accordance with NFPA 70.

G. Grounding and Bonding Requirements, in Addition to Requirements of Section 260526:
   1. Ensure that there is only one AC System bonding connection between grounding system and grounded/neutral conductor, including external connections and connections internal to equipment.

H. Identification Requirements, in Addition to Those Specified in Section 260553:
   1. Use identification nameplate or means of identification acceptable to authority having jurisdiction to identify the presence of multiple power sources and the location of main service disconnecting means and each photovoltaic system disconnecting means. Locate at main service disconnecting means and at each photovoltaic system disconnecting means. Verify format and descriptions with authorities having jurisdiction.
   2. Use identification nameplate to identify each photovoltaic system disconnecting means with text "PV SYSTEM DISCONNECT".
   3. Use identification nameplate or identification label to identify systems equipped with rapid shutdown and associated rapid shutdown switch(es). Format, descriptions, and locations to comply with NFPA 70 and requirements of authorities having jurisdiction.
   4. Use identification nameplate or identification label to identify the information required by NFPA 70 for marking of direct-current photovoltaic power sources. Locate at each DC disconnect means requiring marking.
   5. Use identification nameplate or identification label to identify the interactive system point of interconnection at the disconnecting means as a power source and with the rated AC output current and the nominal operating AC voltage.
   6. Use warning labels to identify electrical hazards for photovoltaic system disconnecting means. Include the word message "Warning - Electric Shock Hazard; Terminals on the line and load sides may be energized in the open position" or approved equivalent.
   7. Use wire and cable markers to identify photovoltaic system source, output, and inverter circuit conductors at all points of termination, connection, and splices.
   8. Use voltage markers, identification labels, stenciled text, or suitable permanent marking approved by authority having jurisdiction to identify exposed raceways, cable trays, pull boxes, junction boxes, and conduit bodies with the text "Warning: Photovoltaic Power Source" at maximum intervals of 10 feet in accordance with NFPA 70.

3.03 SYSTEM STARTUP

A. Obtain Owner's approval prior to performing system startup.

B. Prepare and start system in accordance with manufacturer's instructions.

3.04 CLOSEOUT ACTIVITIES

A. Demonstration: Demonstrate proper operation of system to Owner, and correct deficiencies or make adjustments as directed.

B. Training: Train Owner's personnel on operation, adjustment, and maintenance of photovoltaic system.
   1. Use operation and maintenance manual as training reference, supplemented with additional training materials as required.

2. Provide minimum of two hours of training.

END OF SECTION